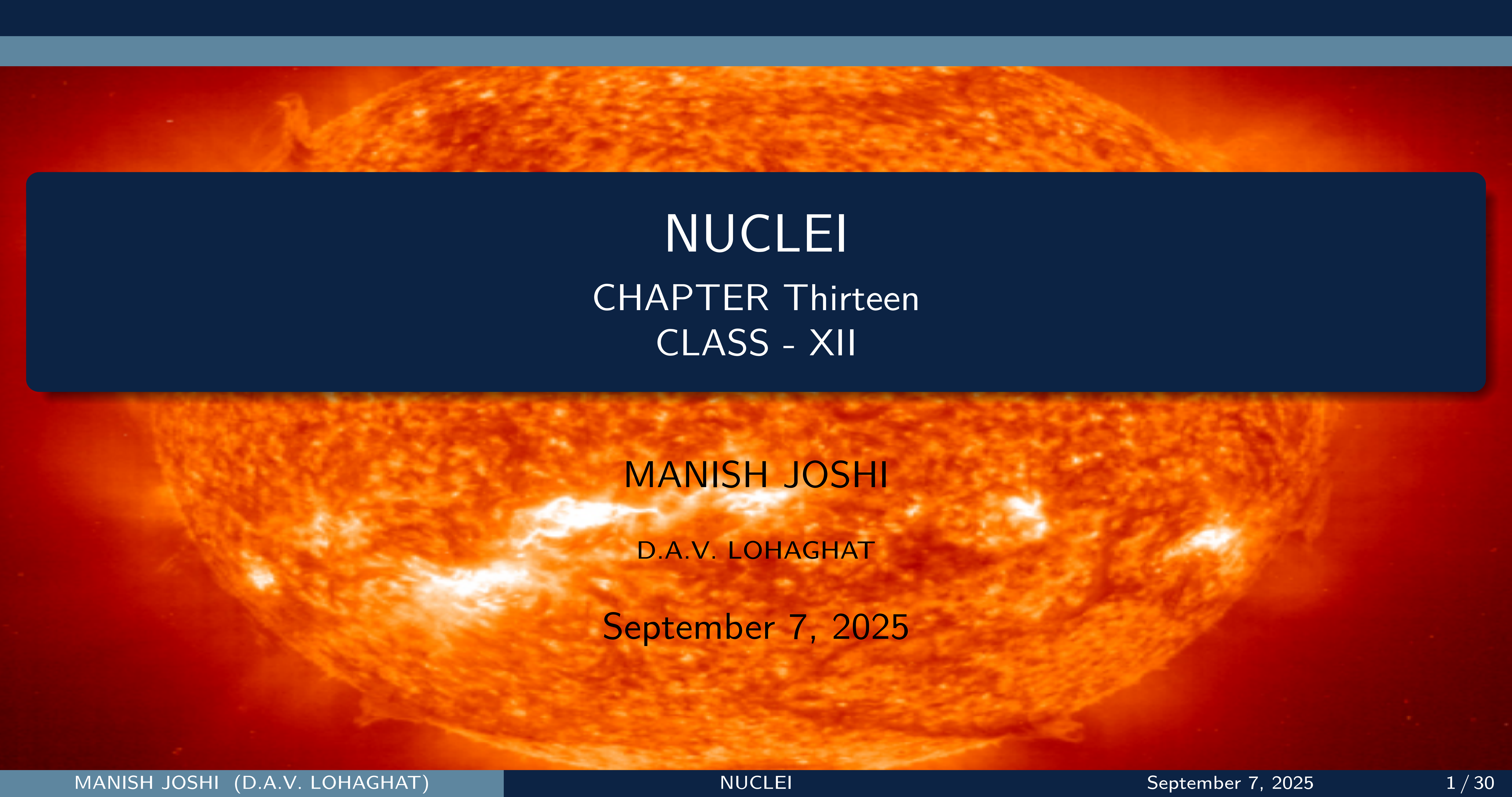




# NUCLEI

## CHAPTER Thirteen

### CLASS - XII

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## Learning Objectives:

- Composition and size of nucleus, nuclear force
- Mass-energy relation, mass defect; binding energy per nucleon and its variation with mass number;
- Nuclear Fission, Nuclear Fusion.



# Introduction

- In every atom, the **positive charge and mass** are **densely concentrated at the centre of the atom** forming its **nucleus**.
- the **radius of a nucleus** was smaller than the radius of an atom by a factor of about  **$10^4$** .
- This means the **volume of a nucleus** is about  **$10^{-12}$**  times the **volume of the atom**. In other words, an **atom is almost empty**.
- Nevertheless, the **nucleus contains most** (more than **99.9%**) of the **mass of an atom**.



# ATOMIC MASSES AND COMPOSITION OF NUCLEUS

## Atomic masses:

- The **mass of an atom is very small**, compared to a kilogram; for example, the **mass of a carbon atom,  $^{12}\text{C}$ , is  $1.992647 \times 10^{-26}$  kg.**
- **Kilogram is not a very convenient unit** to measure such small quantities.
- Therefore, **atomic mass unit (u)**, is **defined as 1/12th of the mass of the carbon ( $^{12}\text{C}$ ) atom.** According to this definition

$$1u = \frac{\text{mass of one } ^{12}\text{C atom}}{12} = \frac{1.992647 \times 10^{-26}}{12} = 1.660539 \times 10^{-27} \text{ kg}$$

- **Accurate measurement of atomic masses** is carried out with a **mass spectrometer.**



# ATOMIC MASSES AND COMPOSITION OF NUCLEUS

## Atomic masses:

- mass of proton = mass of H atom - mass of electron  
$$m_p = m_H - m_e = 1.00783 \text{ u} - 0.00055 \text{ u} = 1.00727 \text{ u} = 1.67262 \times 10^{-27} \text{ kg}$$
- It was earlier thought that the nucleus may contain electrons, but this was ruled out later using arguments based on quantum theory. **All the electrons of an atom are outside the nucleus.**
- The **number of protons** in the nucleus of the atom is **Z**, the **atomic number**.
- The **charge of the nucleus** is **(+Ze)**.
- since the atom is neutral total **charge of the atomic electrons** is thus **(-Ze)**



# ATOMIC MASSES AND COMPOSITION OF NUCLEUS

## Discovery of Neutron:

- Since the **nuclei of deuterium and tritium are isotopes of hydrogen**, they must **contain only one proton** each. But the **masses of the nuclei of hydrogen, deuterium and tritium** are in the **ratio of 1:2:3**. Therefore, the nuclei of deuterium and tritium **must contain**, in addition to a proton, some **neutral matter**.
- This hypothesis was verified in 1932 by **James Chadwick** who **observed emission of neutral radiation** when **beryllium nuclei were bombarded with alpha-particles**



# ATOMIC MASSES AND COMPOSITION OF NUCLEUS

## Discovery of Neutron:

- Chadwick assume that the neutral radiation consists of a new type of neutral particles called **neutrons**. From conservation of energy and momentum, he was able to determine the **mass of new particle as very nearly the same as mass of proton**
- mass of neutron  $m_n = 1.00866u = 1.6749 \times 10^{-27} \text{ kg}$



# ATOMIC MASSES AND COMPOSITION OF NUCLEUS

## ISOTOPES

- **Isotopes:** Atoms of same element (same  $Z$ ) but different  $A$ .
- Example: Hydrogen has three isotopes:



- Chlorine has isotopes  $34.98u$  (75.4%) and  $36.98u$  (24.6%). Average atomic mass =  $35.47u$ .
- Chemical properties of isotopes are identical (same electronic configuration).



# ATOMIC MASSES AND COMPOSITION OF NUCLEUS

## COMPOSITION OF NUCLEUS

- The composition of a nucleus can now be described using the following terms and symbols:
- **$Z$  = atomic number = number of protons**
- **$N$  = number of neutrons**
- **$A$  = mass number =  $Z + N$  = total no. of proton and neutron (or nucleons)**
- Nuclear species or nuclides are shown by the notation  ${}^A_ZX$  where  $X$  is the chemical symbol of the species.
- For example, the nucleus of gold is denoted by  ${}^{197}_{79}Au$ . It contains 197 nucleons, of which 79 are protons and the rest 118 are neutrons.



# ATOMIC MASSES AND COMPOSITION OF NUCLEUS

## ISOBARS AND ISOTONES

- **Isobars:** Same mass number ( $A$ ), different  $Z$ . Example:  ${}^3_1\text{H}$  and  ${}^3_2\text{He}$ .
- **Isotones:** Same neutron number ( $N$ ), different  $Z$ . Example:  ${}^{198}_{80}\text{Hg}$  and  ${}^{197}_{79}\text{Au}$ .

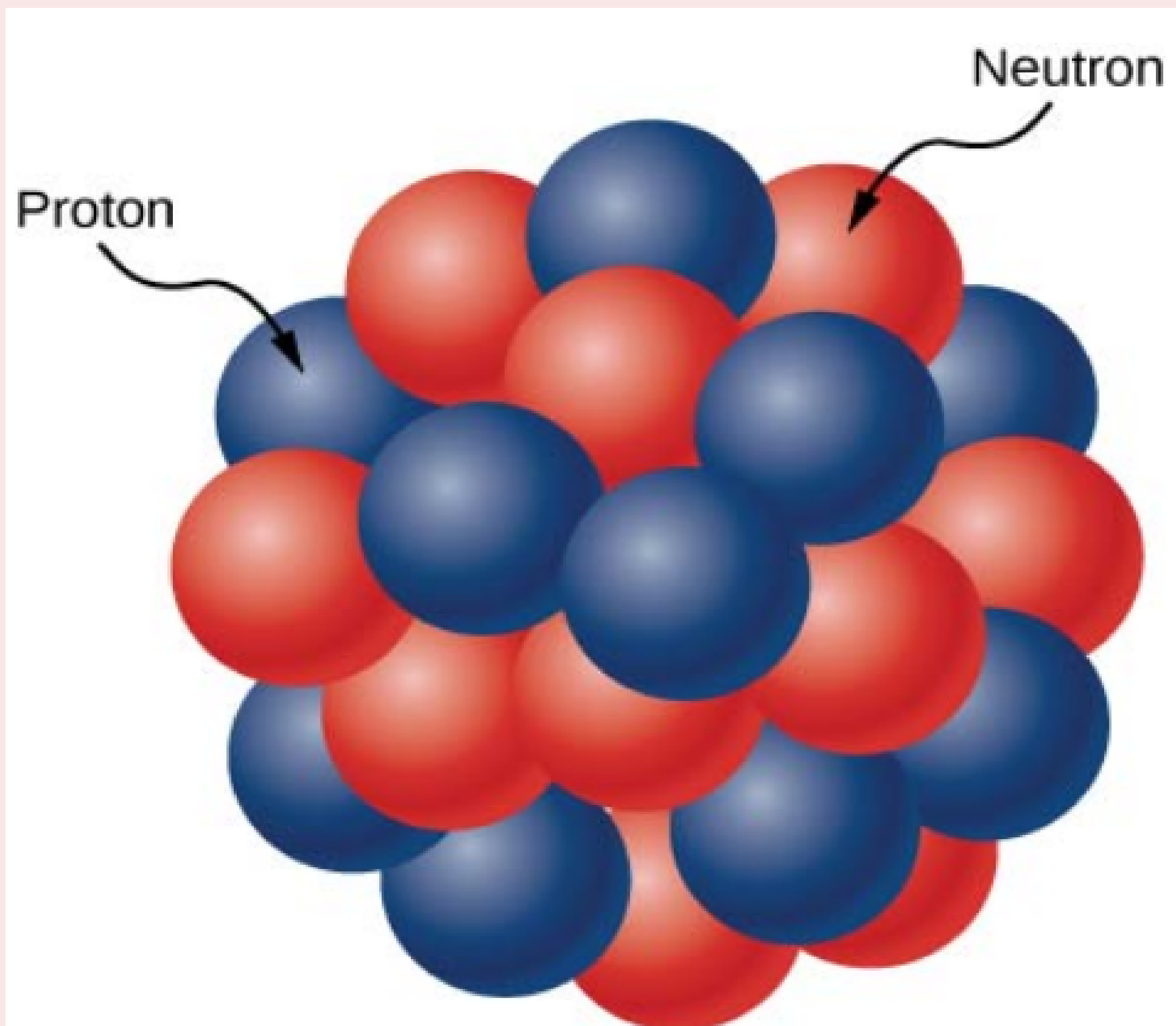


# SIZE OF THE NUCLEUS

- Rutherford's suggestion, Geiger and Marsden performed their classic experiment: on the scattering of  $\alpha$ -particles from thin gold foils. Their experiments revealed that the distance of closest approach to a gold nucleus of an  $\alpha$ -particle of kinetic energy 5.5 MeV is about  $4.0 \times 10^{-14}$  m.
- the **actual size of the nucleus has to be less than  $4.0 \times 10^{-14}$  m.**
- By performing scattering experiments in which **fast electrons**, instead of  $\alpha$ -particles, are projectiles that bombard targets made up of various elements, the **sizes of nuclei of various elements have been accurately measured.**



# SIZE OF THE NUCLEUS



- The simplest model of the nucleus is a densely packed sphere of nucleons. **The volume  $V$  of the nucleus** is therefore **proportional to the number of nucleons  $A$** , expressed by

$$V = \frac{4}{3}\pi R^3 = kA$$

- where  **$R$  is the radius of a nucleus** and  **$k$  is a constant** with units of volume. Solving for  $R$ , we have

$$R = R_0 A^{1/3}$$

- where  **$R_0 = 1.2 \times 10^{-15} \text{ m} (= 1.2 \text{ fm})$** .



# DENSITY OF THE NUCLEUS

- $\text{Density} = \frac{\text{Mass of nucleus}}{\text{Volume of nucleus}} = \frac{A}{\frac{4}{3}\pi R^3} = \frac{A}{\frac{4}{3}\pi R_0^3 A}$
- $\text{Density} = \frac{1}{\frac{4}{3}\pi R_0^3} = \text{Constant} = 2.3 \times 10^{17} \text{ kgm}^{-3}$



# Example 1

Given the **mass** of iron nucleus as **55.85 u** and **A=56**, find the **nuclear density**?  
( $1u = 9.27 \times 10^{-26} \text{ kg}$ )



# Mass – Energy

- **Einstein** showed from his **theory of special relativity** that it is necessary to **treat mass as another form of energy**.
- **Before** the advent of this **theory of special relativity** it was **presumed that mass and energy were conserved separately** in a reaction.
- However, **Einstein** showed that **mass is another form of energy** and **one can convert mass-energy into other forms of energy**, say kinetic energy and vice-versa.
- Einstein gave the famous **mass-energy equivalence relation**

$$E = mc^2$$

Here,  $m$  is energy equivalent of mass and  $c$  is speed of light in vacuum.



## Example 2

Calculate the energy equivalent of 1 g of substance.



# Mass Defect

- We have seen that the **nucleus is made up of neutrons and protons.**
- Therefore **it may be expected** that the **mass of the nucleus is equal to the total mass of its individual protons and neutrons.**
- However, According to nuclear particle experiments, the **total mass of a nucleus ( $M_{nuc}$ ) is less than the sum of the masses of its constituent nucleons (protons and neutrons).**



# Mass Defect

- For example, let us consider  ${}^{16}_8\text{O}$  ; a nucleus which has **8 neutrons and 8 protons**.
- Mass of 8 neutrons  $= 8 \times 1.00866 \text{ u}$
- Mass of 8 protons  $= 8 \times 1.00727 \text{ u}$
- Mass of 8 electrons  $= 8 \times 0.00055 \text{ u}$
- Therefore the **expected mass  ${}^{16}_8\text{O}$  nucleus**  $= 8 \times 2.01593 \text{ u} = 16.12744 \text{ u}$ .
- The **atomic mass** of  ${}^{16}_8\text{O}$  found from **mass spectroscopy** experiments is seen to be **15.99493 u**. Subtracting the mass of 8 electrons ( $8 \times 0.00055 \text{ u}$ ) from this, we get the **experimental mass** of  ${}^{16}_8\text{O}$  nucleus to be **15.99053 u**.
- Thus, we find that the **mass of the  ${}^{16}_8\text{O}$  nucleus is less than the total mass of its constituents by 0.13691 u**.



# Mass Defect

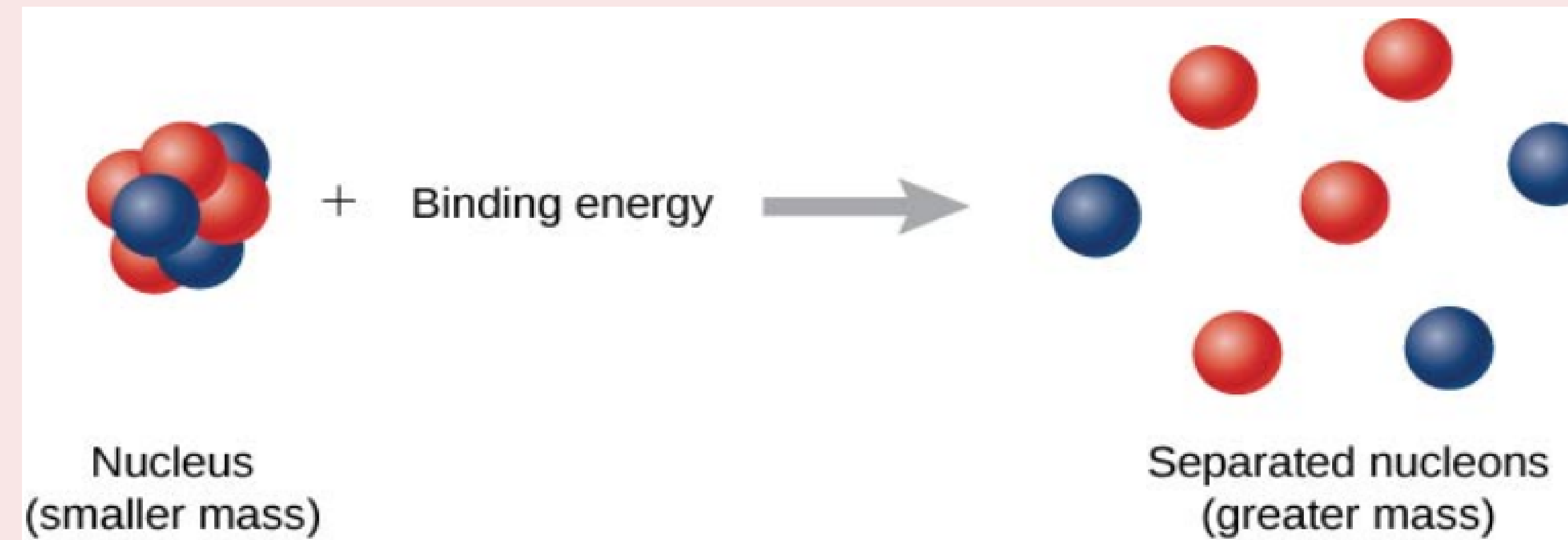
- The **difference in mass of a nucleus and its constituents,  $\Delta M$ , is called the mass defect**, and is given by

$$\Delta M = [Z M_p + (A - Z) M_n] - M_{nucl}$$

- where  $Z M_p$  is the **total mass of the protons**,  $(A - Z) M_n$  is the **total mass of the neutrons**, and  $M_{nuc}$  is the **mass of the nucleus**.



# Binding Energy



- The **formation of a nucleus** from a system of isolated protons and neutrons is therefore an **exothermic reaction**-meaning that it **releases energy**. The **energy emitted**, or radiated, in this process is  $(\Delta M)c^2$ .
- Now **imagine this process occurs in reverse**. Instead of forming a nucleus, **energy is put into the system to break apart the nucleus**. The amount of energy required is called the **total binding energy (BE)**, ( $E_b = \Delta Mc^2$ ).



**Example 13.3** Find the energy equivalent of one atomic mass unit, first in Joules and then in MeV. Using this, express the mass defect of  $^{16}_8\text{O}$  in  $\text{MeV}/c^2$ .

**Solution**

$$1\text{u} = 1.6605 \times 10^{-27} \text{ kg}$$

To convert it into energy units, we multiply it by  $c^2$  and find that

$$\begin{aligned} \text{energy equivalent} &= 1.6605 \times 10^{-27} \times (2.9979 \times 10^8)^2 \text{ kg m}^2/\text{s}^2 \\ &= 1.4924 \times 10^{-10} \text{ J} \end{aligned}$$

$$= \frac{1.4924 \times 10^{-10}}{1.602 \times 10^{-19}} \text{ eV}$$

$$= 0.9315 \times 10^9 \text{ eV}$$

$$= 931.5 \text{ MeV}$$

$$\text{or, } 1\text{u} = 931.5 \text{ MeV}/c^2$$

$$\begin{aligned} \text{For } ^{16}_8\text{O}, \quad \Delta M &= 0.13691 \text{ u} = 0.13691 \times 931.5 \text{ MeV}/c^2 \\ &= 127.5 \text{ MeV}/c^2 \end{aligned}$$

The energy needed to separate  $^{16}_8\text{O}$  into its constituents is thus  $127.5 \text{ MeV}/c^2$ .



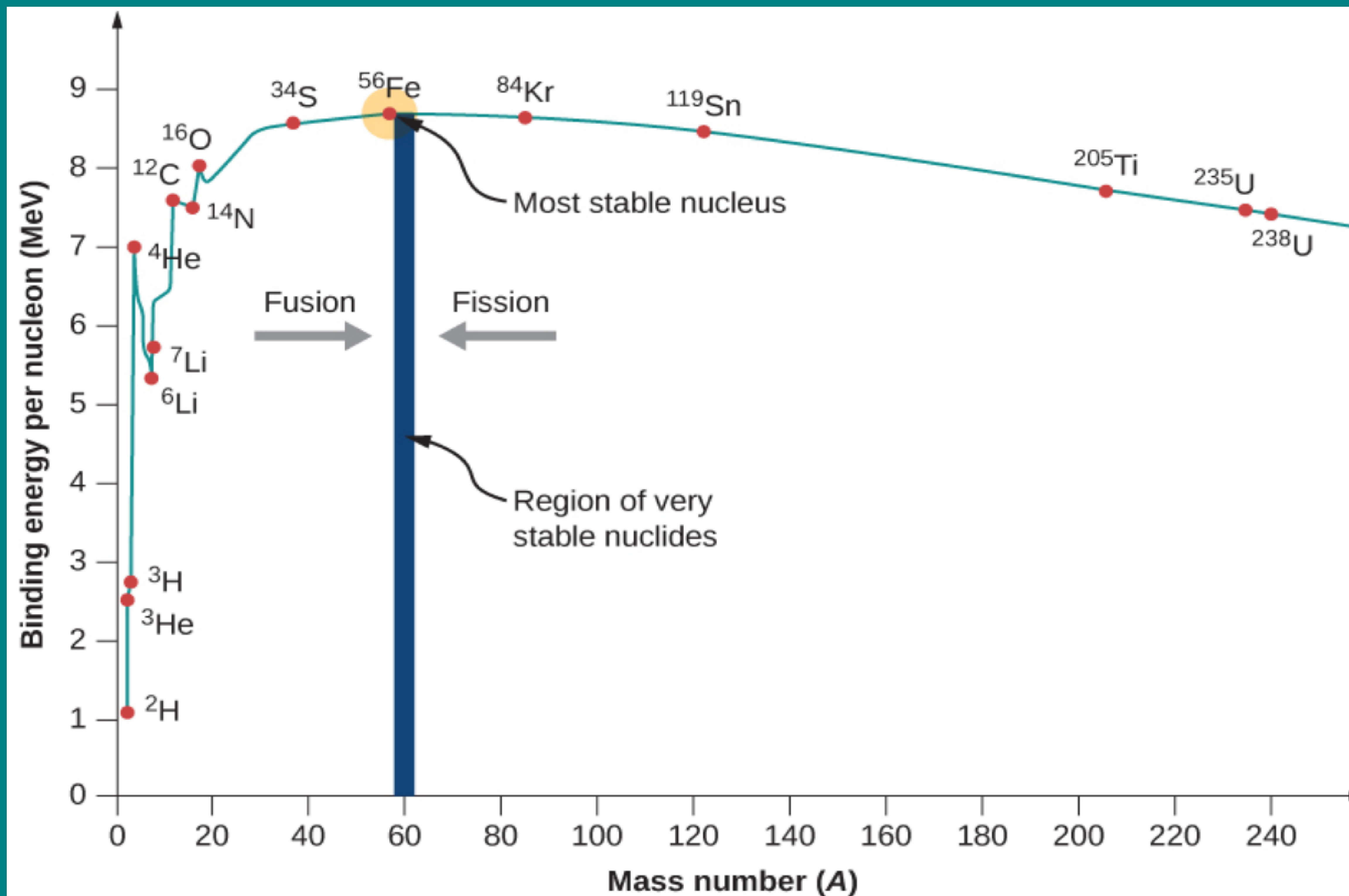
# Graph of Binding Energy per Nucleon

- In nuclear physics, one of the most important experimental quantities is the binding energy per nucleon ( $E_{bn}$ ), which is the **ratio of the binding energy**  $E_b$  of a nucleus **to the number of the nucleons**,  $A$ , in that nucleus:

$$E_{bn} = \frac{E_b}{A}$$

- We can think of **Binding energy per nucleon** as the **average energy per nucleon needed to separate a nucleus into its individual nucleons**.
- If the **Binding energy per nucleon** is relatively **large**, the **nucleus is relatively stable**.

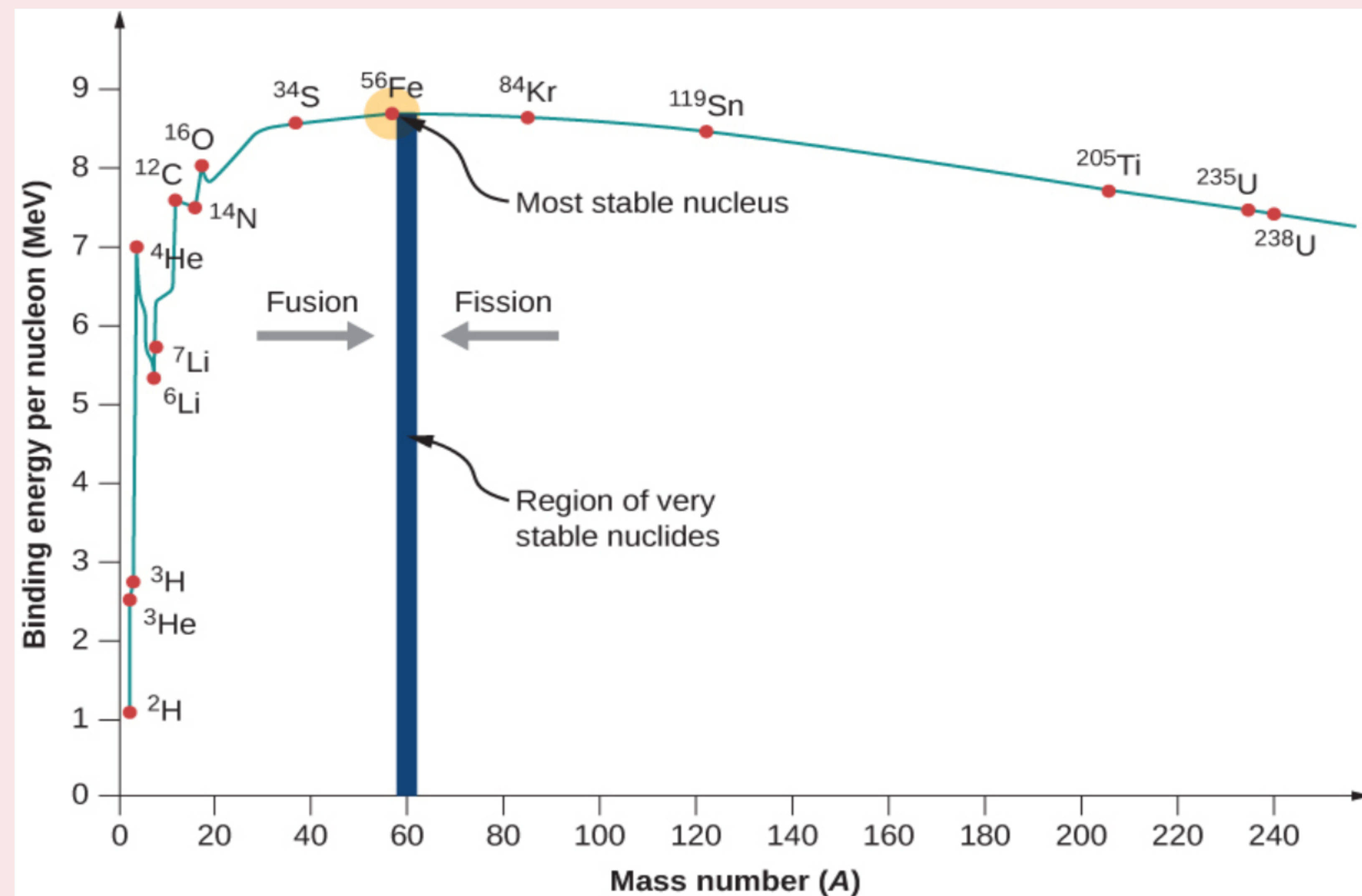




**Figure 10.7** In this graph of binding energy per nucleon for stable nuclei, the BEN is greatest for nuclei with a mass near  $^{56}\text{Fe}$ . Therefore, fusion of nuclei with mass numbers much less than that of Fe, and fission of nuclei with mass numbers greater than that of Fe, are exothermic processes.



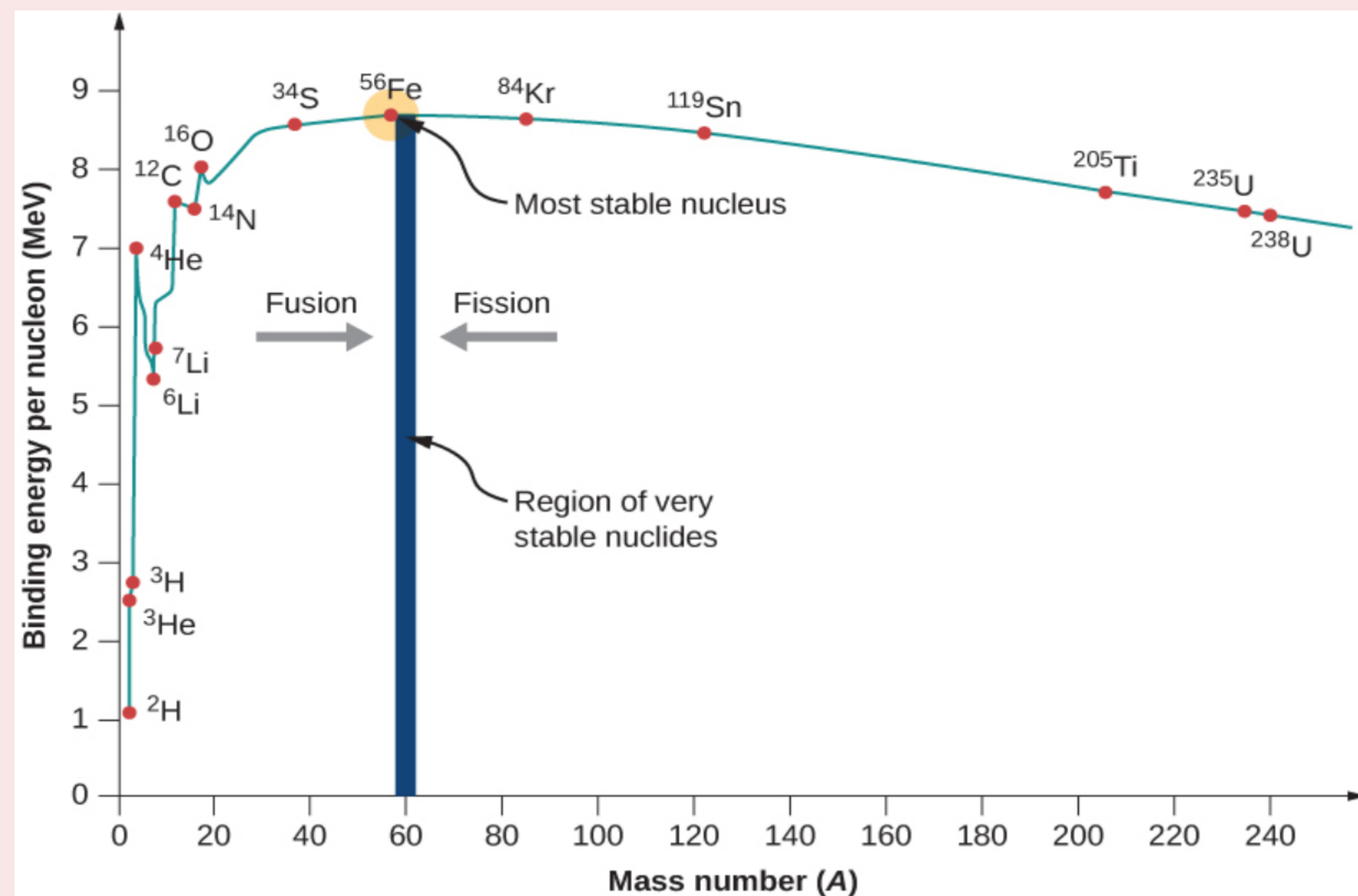
# Graph of Binding Energy per Nucleon



- We notice the following **main features of the plot**:
- the **binding energy per nucleon**,  $E_{bn}$ , is practically **constant**, i.e. practically independent of the atomic number of middle mass number nuclei ( $30 < A < 170$ ).
- The curve has a **maximum** of about 8.75 MeV **for Fe**  $A = 56$  and has a value of 7.6 MeV
- $E_{bn}$  is **lower** for both **light nuclei** ( $A < 30$ ) and **heavy nuclei** ( $A > 170$ ).



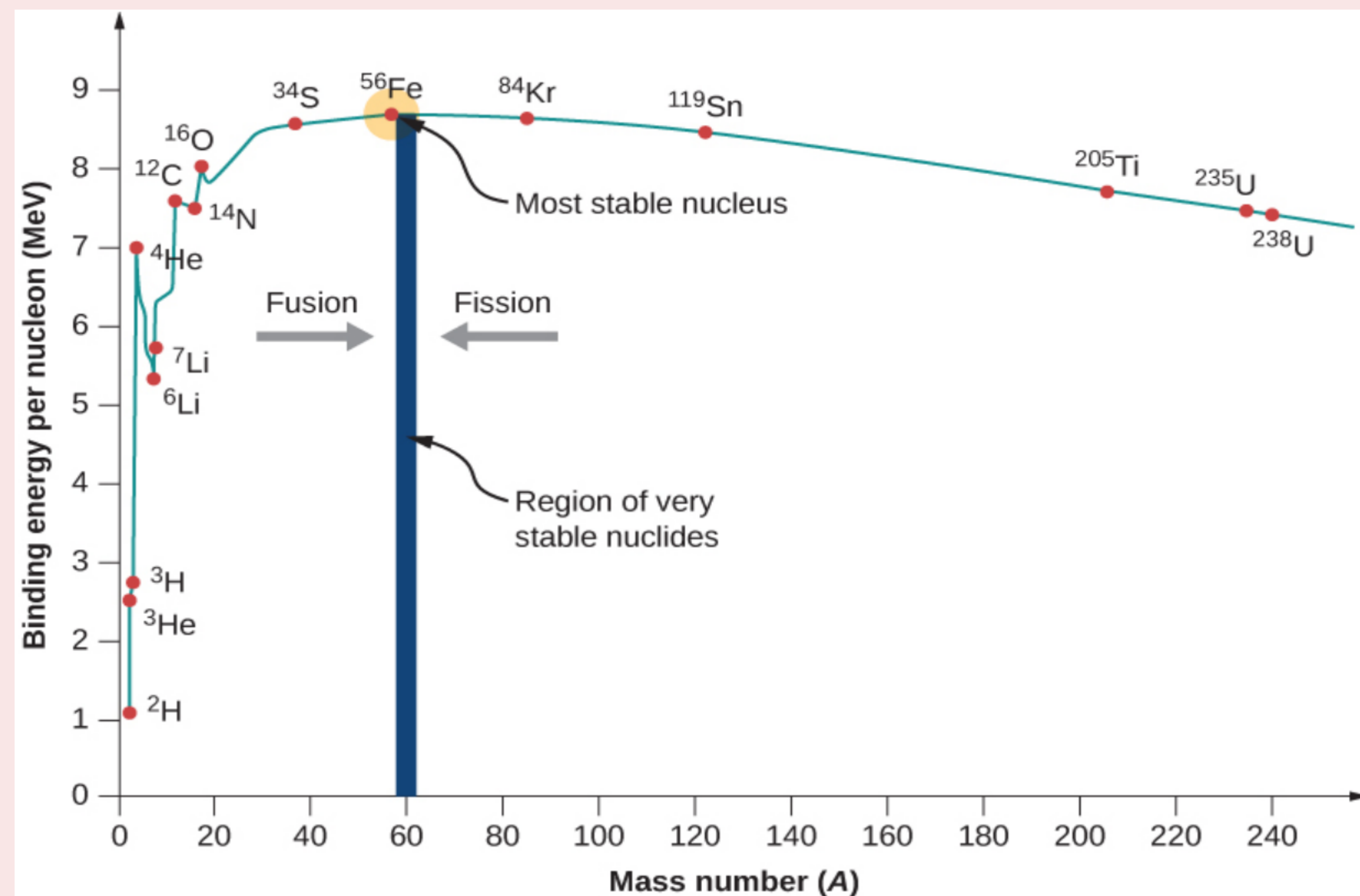
# Graph of Binding Energy per Nucleon



- typical  $E_{bn}$  values range from 6-10 MeV, with an average value of about 8 MeV.
- In other words, it takes **several million electron volts to pry a nucleon** from a typical nucleus, as **compared to just 13.6 eV to ionize an electron** in the ground state of hydrogen.
- This is why **nuclear force is referred to as the "strong" nuclear force.**



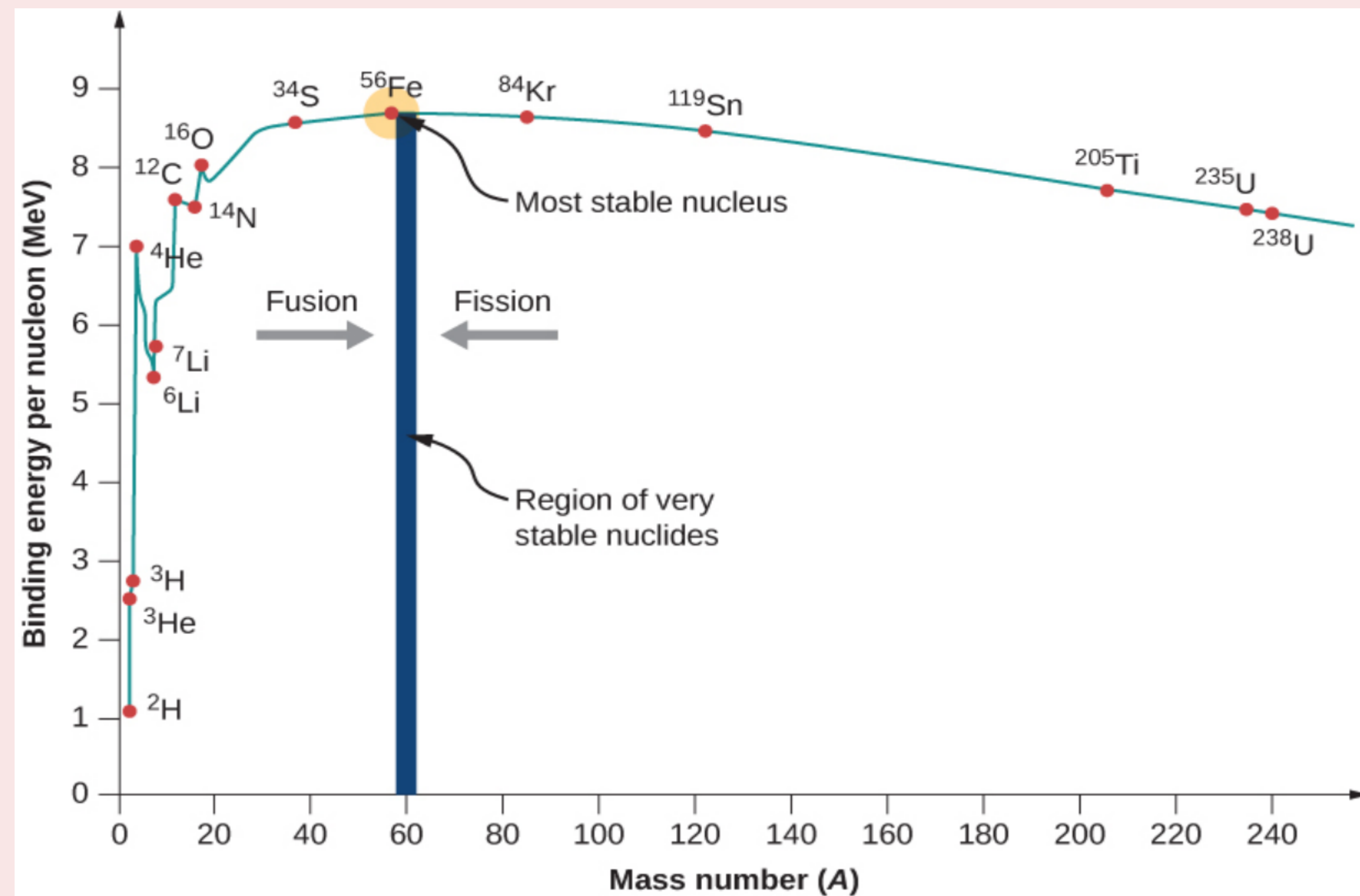
# Graph of Binding Energy per Nucleon



- The graph rises at low  $A$ , peaks very near iron ( $\text{Fe}$ ,  $A = 56$ ), and then tapers off at high  $A$ .
- The peak value suggests that the iron nucleus is the most stable nucleus in nature (it is also why nuclear fusion in the cores of stars ends with  $\text{Fe}$ ).
- The reason the graph rises and tapers off has to do with **competing forces in the nucleus**.



# Graph of Binding Energy per Nucleon



- At low values of  $A$ , attractive nuclear forces between nucleons **dominate over repulsive electrostatic forces** between protons.
- But at **high values of  $A$** , **repulsive electrostatic forces** between forces **begin to dominate**, and **these forces tend to break apart the nucleus** rather than hold it together.



# NUCLEAR FORCE

## Why a new force is needed?

- Binding energy per nucleon  $\approx 8$  MeV, much larger than atomic binding energies (eV range).
- Coulomb repulsion between protons cannot be balanced by gravity (too weak).
- A new strong attractive force is required to hold nucleons together inside nucleus.



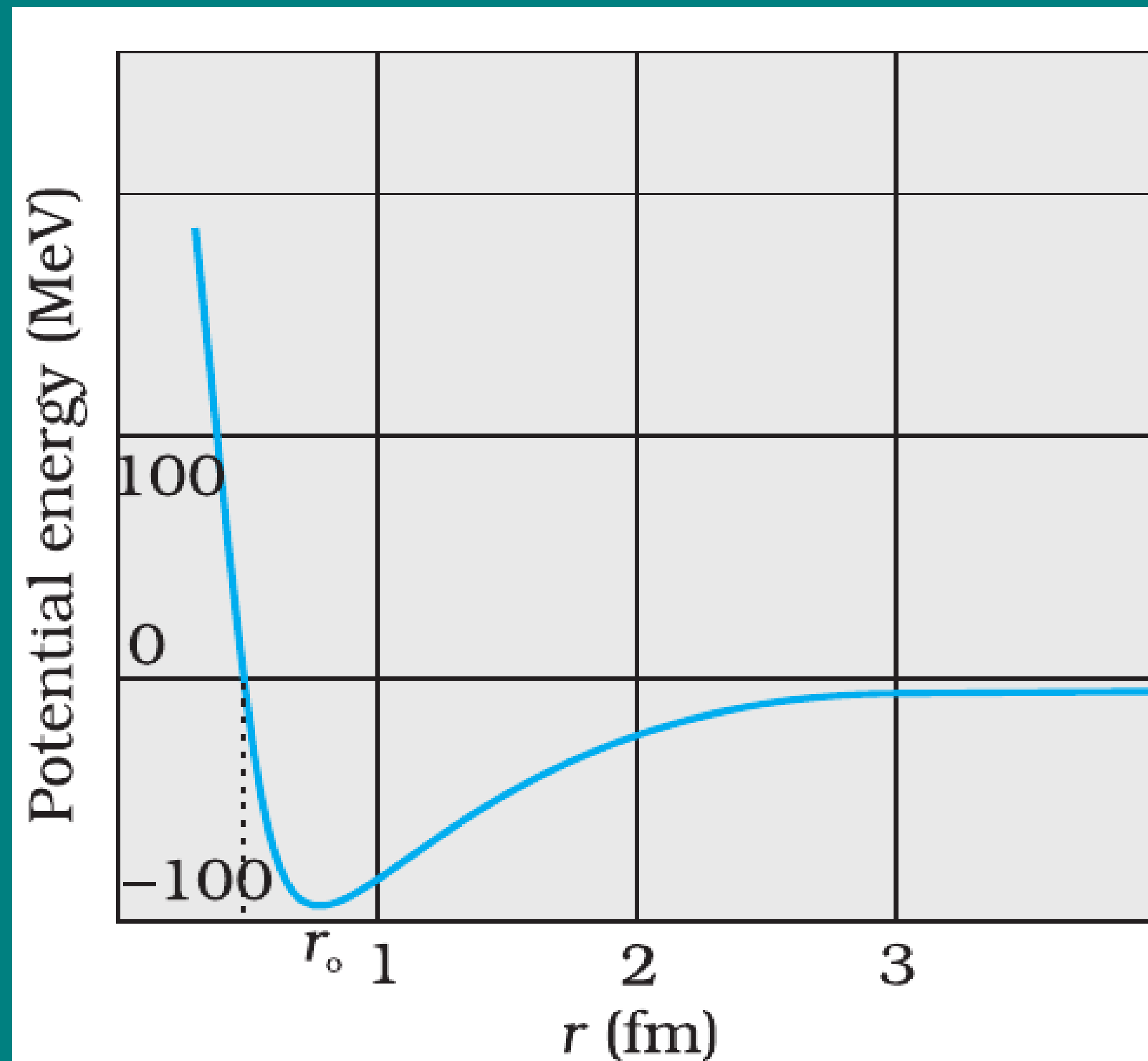
# CHARACTERISTICS OF NUCLEAR FORCE

## Main Properties:

- **Very Strong:** Stronger than both Coulomb and gravitational forces.
- **Short Range:** Effective only up to a few femtometres (fm); falls rapidly beyond.
- **Saturation Property:** Each nucleon interacts only with its nearest neighbours.
- **Charge Independent:** Same for p–p, n–n, and p–n interactions.
- **Attractive–Repulsive Nature:** Attractive beyond 0.8 fm; strongly repulsive below 0.8 fm.
- **No simple formula:** Unlike Coulomb's or Newton's law, its exact mathematical form is unknown.



# POTENTIAL ENERGY CURVE



**FIGURE 13.2** Potential energy of a pair of nucleons as a function of their separation. For a separation greater than  $r_0$ , the force is attractive and for separations less than  $r_0$ , the force is strongly repulsive.

## Nuclear Potential Energy vs Distance

- Potential energy minimum at  $\approx 0.8$  fm  $\Rightarrow$  stable bound state.
- Attractive for  $r > 0.8$  fm.
- Strongly repulsive for  $r < 0.8$  fm.